

Optical Tweezers: Methods for evaluating trap stiffness

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An optical trap is a laser configuration that allows for the trapping and driving of particles. This paper explores two methods for evaluating optical trap stiffness governed by variable α , essential in calculating spring like forces the trap exerts on targets suspended in a medium. This α , combined with a calibration from a quadrant photodiode (QPD), measures the effective displacement of the trapped object in space. Furthermore, knowledge of the viscosity of the suspension medium and scale of force involved is used to describe the trap mechanics. Potential pitfalls in the methods are described as encountered, as well as a series of results for the example trap of $\lambda = 975\text{nm}$ on a 3 micron bead.

I. BACKGROUND

The optical trap, or optical tweezers as it is also known, has been a critical tool in the development of biophysics, where measuring the pico-Newton-scale forces at the micrometer scale is key to understanding the mechanics behind organelles, DNA, bacteria, and other microorganisms. The trap we will discuss was first invented in the 1980's following a series of seminal research papers that Ashkin Et al. published from the dawn of the 1970's [1][2]. The initial discovery that led to the optical trap was that particles caught in the path of a laser beam would not only experience a scattering force along the direction of beam propagation, much like a ball caught in the blast of a fire-hose, but that they would also experience what was dubbed a gradient force axially inward toward the center of the beam. This led to the development of the optical trap, where the beam is focused to a point where the combination forces admit to a stable trapping of particles positioned near the focal point.

The mechanics of the trapping forces can be difficult to describe in a both concise and accurate way due to the scales involved. The two typical regimes in describing optics are typically when the wavelength of the light is much smaller than the scale of the object it is interacting with (Ray optics), or when the wavelength is much larger than the scale of the object (Rayleigh Scattering). For optical traps, we typically have the wavelength around $\lambda = 970\text{nm}$, which is approximately a micron. This happens to be the scale of our objects of interest. Because of this, neither model really describes well what is happening physically, but for the purpose of building intuition we will describe the ray optics mechanics of the trap, which owe the gradient force to the changing and imparting of momentum from photons refracting through the trapped object. If one were to model this under Rayleigh scattering, one would consider the bead as a dielectric sphere, upon which

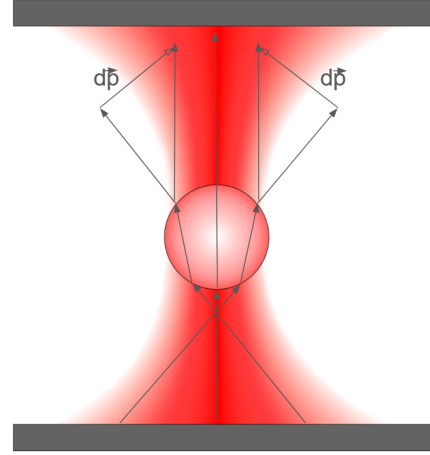


FIG. 1. Ray optics model: The change in momentum of photons $d\vec{p}$ in combination with scattering force keeps the bead centered in the trap. Movement away from the center experiences a restoring force due to the difference in intensity decreasing off center.

dipole charge is induced from the interaction with the electromagnetic photons. Both of these methods while not necessarily completely accurate, provide models that for the purpose of using the trap are perfectly adequate.

Laser light is focused through a condenser lens, much like light in a microscope, such that the focal point of the beam is centered at the site of the target to be trapped. The target, in our discussion a $3.12\mu\text{m}$ diameter bead, will interact with the beam and be subject to trapping forces. Following the interaction with the bead, the scattered and unscattered laser light is passed through a lens on the other side. This is then directed towards a quadrant photodiode sensor (QPD). The beam, which now has an interference pattern due to the overlapping of phase shifted light picked up by scattered photons, is recorded. Based on the position of the bead, the center of the distribution of light intensity will shift. This allows us to see relative position changes of the bead in the trap, but requires calibrating against a fixed bead to translate the voltage readout of the sensor to actual position measure-

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ments. The shift in the interference pattern with bead movement depends both on the transmissive properties of the bead, as well as the size/geometry, so calibration of the QPD signal must be done on objects that represent as closely as possible the target in question. In this paper we will focus on 3 micron beads, but those looking to study say E.Coli bacteria would carry out this procedure on 1 micron beads to match the scale of the bacterium.

II. MODELING MECHANICS

Considering that the restoring force increases with displacement from the trap center, we can approximate the force, like many things in physics, as a harmonic potential. We consider the one dimensional case.¹

$$V_{trap}(x) = \frac{1}{2}\alpha\langle x^2 \rangle \quad (1)$$

Then, according to the equipartition theorem from statistical mechanics, we can equate this potential to Boltzmanns constant and Temperature for the energy of a trapped bead at thermal equilibrium. This permits us an understanding of the mechanics from an energetic perspective.

$$\frac{1}{2}\alpha\langle x^2 \rangle = \frac{1}{2}k_B T \quad (2)$$

This spring like, Hookean model $F(x) = -kx = \alpha x$ for the trap will be valid for small displacements. Next, we must understand the mechanics of the targets in their respective medium. At micron scales, targets in water experience a comparatively large amount of viscous drag compared to the typical forces they encounter. The trap itself will also apply just pico-newtons of force. To understand how to model this, we first consider the general Newtonian force equation governed by [4] which combines acceleration, our drag force as governed by Stokes, and the restoring force provided by the trap for displacements away from the trap center.

$$\sum f(t) = F_{Newtonian}(x) + F_{drag}(x) + F_{restoring}(x) \quad (3)$$

$$\sum f(t) = m\ddot{x}(t) + \beta\dot{x}(t) + \alpha x(t) = F(t) \quad (4)$$

In fact, the drag is so high that the acceleration is considered to be zero, as terminal velocity is reached almost immediately from any applied force. This is commonly described as an over damped system. With this, our force equation reduces to a combination of the drag and trap force. [5]

$$F(t) = \beta\dot{x}(t) + \alpha x(t) \quad (5)$$

Variable β is governed by Stokes drag equation [6], and α represents our traps spring constant. For Stokes drag on a sphere, μ is the medium and d is the diameter of the particle. These are both known or measurable quantities.

$$F_{drag}(t) = -3\pi\mu d\dot{x}(t) \quad (6)$$

This leaves us with a need to measure α to determine the force within the trap, given the positions and velocities. We can use [1] to determine α given k_b and T , but in the absence of knowing a good value of k_B we can also record the movement of the target over time in the trap. Due to thermal excitations of molecules in the medium, the bead in the trap will be constantly bombarded from all sides by random forces. These random forces should be random, and so Einstein proposed that the power spectrum associated with this random force induced movement should look like white noise. Random movement of particles suspended in a medium like this is known as Brownian motion, and it is something we can measure with the QPD, and consequently acquire not only a value for trap stiffness but also a value for Boltzmanns constant.

We will now describe the apparatus used to determine α , including a measurement of Boltzmanns constant.

III. EXPERIMENTAL APPARATUS

The optical trap and associated measurement apparatus is described in the diagram shown below [2]. It is fundamentally an inverted microscope with large numerical aperture oil immersion lenses, where a 975nm laser with a width that overfills the back of the condenser lens is passed through the device alongside the visible light through the use of hot mirrors. As mentioned, the laser forms the trap and then passes to the QPD for position measurement. The light for the microscope that is passed through the slide is routed in the opposite direction, from the slide to a CCD imaging sensor. CCD imaging for methods in this paper is primarily concerned with finding beads to put in the trap within the medium. These images/videos can also be processed for analysis if desired, though the QPD is recommended for any data desired at a sampling rate above 30Hz due to CCD limitations.

Samples to place into the trap are prepared on glass slides. Strips of double sided tape are used between the slide and the cover slip to form a channel, in which the solution containing the colloid bead suspension is placed. The channel is then sealed at the ends with clear nail varnish and allowed to dry (but not dry out!) before being placed onto the stage ready for trapping.

Slides are mounted on a movable stage provided by ThorLabs. The stage is electronic and can be controlled in x, y, and z axis by the use of the attached micrometers, but can also be driven electronically through the

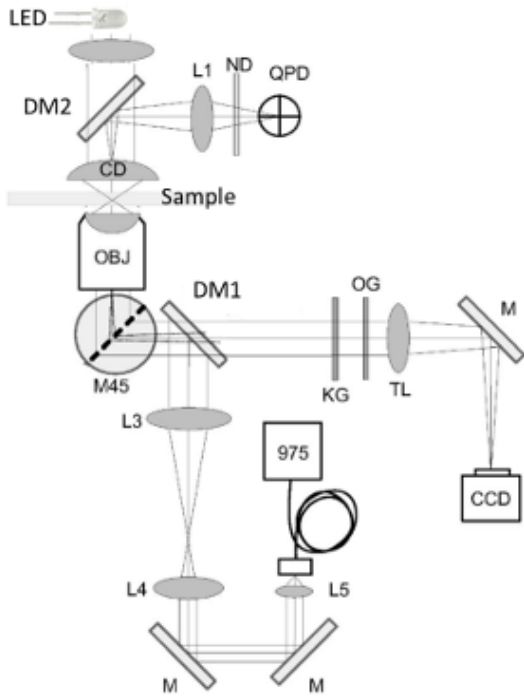


FIG. 2. Schematic diagram of the optical beam paths and apparatus. Source:Jlab manual

internal piezos. The same electronics are used to measure the stage displacement via the piezo strain. To ensure more absolute measurements associated with piezo strain, there is a capacitive feedback loop.

OKTB is the software of choice for outputting data. It is run through matlab and receives the QPD and strain gauge position data from the apparatus. We will use this software to program different stage movements to perform measurements described in the data collection subsection.

IV. METHOD

The following two methods are used to get a measurement for alpha. To allow for solid analysis, temperature should be maintained as stable as possible, and sources of background vibration that could pass to the apparatus should be reduced. We had our apparatus atop a simple optical table to help reduce vibration.

IV.1. Data Collection: Brownian Motion

Following our approximation that the bead is overdamped in 5, and assuming that the Brownian motion is a random process, we use the Wiener-Khinchin theorem to define a “power spectral distribution” function (PSD) or “power spectrum” via the Fourier transform of the time-averaged autocorrelation function.

$$S_{xx}(f) = \sqrt{\frac{k_B T}{\pi^2 \beta (f^2 + f_0^2)}} \quad (7)$$

$$f_0 = \frac{\alpha}{2\pi\beta} \quad (8)$$

This derives from the result that the power spectrum of white noise is $\sqrt{4\beta k_B T}$.

The power spectrum above is a Lorentzian distribution, and its behavior can be analyzed and understood looking at the term in parenthesis. At low f , the f_0 term will dominate the denominator, giving a constant line. At high f , the f term will instead dominate and give a different constant line. This leads us to expect a curve in log-log that has two linear regions with an intermediate knee transition. This knee is what we will fit to find f_0 , which will be at the turning point of the power spectrum.

Once the slide is in the stage and positioned in the trap, we set an appropriate height. To ensure an appropriate plane of focus in the z axis (designated as the upwards orientation, opposing the direction of the gravitational force), after a slide of 3.12 μ m bead solution was inserted into the trap, the trap was aligned with edge of the bead channel using the micrometer controls. The edge of the tape was then focused on as a starting reference. Then moving away from the sidewall, an area free of beads or other objects in the slide was chosen to take a flat field image to subtract from the CCD image feed. This allows us to remove artifacts caused by dust and grime on the optical array making it easier to find actual beads.

A bead was then located and centered on the display. The trap was powered on to contain the bead, and we then recorded the power spectrum with intervals of 10s each at 50,000Hz of the trapped bead at varied laser powers. This data was then collected and processed in a jupyter notebook. In hindsight, we should have checked the datasheet for the device as the QPD is only accurate up to 10,000Hz. Given the a simplified interpretation of the Nyquist-Shannon sampling theorem, we are then limited to studying frequencies up to 5,000Hz. For an interesting discussion on Nyquist-Shannon sampling theorem and its application to systems see “Sampling: What Nyquist Didn’t Say, and What to Do About It” [3]

IV.2. XY Scan

We can also study the motion of the bead to determine α . This approach leverages the equipartition theorem [1] and thus a QPD calibration is required to get reliable position data from which to get the expected value of $\langle x^2 \rangle$. Since it relies on QPD calibration, any error in the calibration will also affect the XY scan for alpha, introducing noise through the position squared, so this method is quite susceptible to noise. The approach is the same as the Brownian method, but once the bead

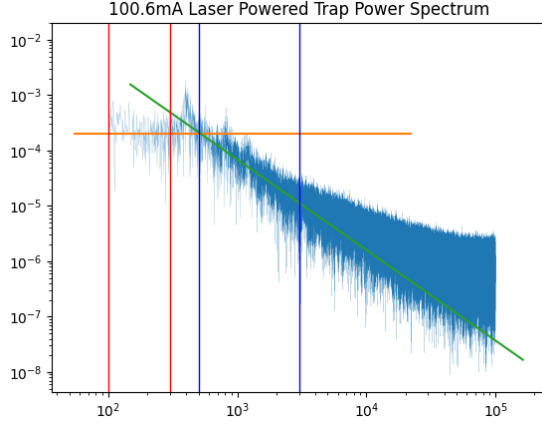


FIG. 3. Power spectra for 100mA powered trap. Vertical lines show the bounds of the windows used for fitting linear curve.

is trapped, the stage is driven via an electronic signal sent to the piezos. The resulting QPD signal can then be transformed with calibration, and for each axis swept we take the variance of the position.

V. ANALYSIS

V.1. Brownian Motion and Boltzmanns Constant

Following data collection and import into a jupyter notebook, the QPD data was then converted into a power spectrum using `scipy.fft.fft`. Since calibration of the qpd is only concerned with the orientation of the QPD axis and the relative amplitude of displacements, it was not necessary for us to analyze this data given we are just seeking the knee frequency of the curve. In my analysis, a simplified curve fitting was performed with a sliding window on both the flat part and the sloped part of the power spectrum curve on a log-log plot to find the knee point and resulting f_0 .

An example of two power spectra at the extremes of our available laser power are presented in [ref]. We have plotted also the f_0 , and resulting trap stiffness constants.

We find through this method a trap stiffness in the range of 0.16-0.21 (pN/nm). This is in line with expected results, but it is worth noting that this method in the way we have conducted it does not consider the separate x and y axis of the trap and assumes homogeneity. A linear fit can be used to find intermediate values between those measured, and given the somewhat non-linear behavior between the extrema this should be fit locally for the desired stiffness in the absence of a non linear model to understand the S shaped curve.

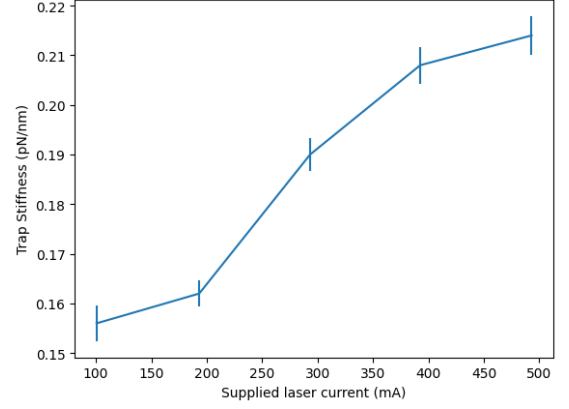


FIG. 4. Calculations of α , trap stiffness, at varied supplied laser currents.

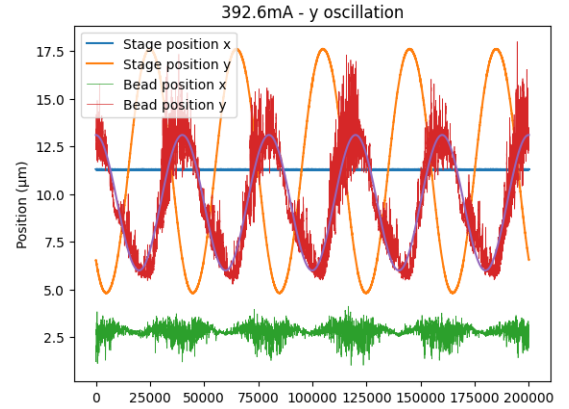


FIG. 5. An example of good Y oscillation data, where the signal in X qpd is minimal and the Y QPD signal is a reduced phase shifted version of the driving signal.

V.2. Measuring α with XY scans

Next, we performed XY scans on the same trapped bead. Given the time scales involved to run these tests we assume the degradation to the bead from the laser to be negligible, something that we could not assume had we used Latex-Polymer beads as Ashkin Et al. did in their original papers.

The trapped bead was swept sinusoidally at a frequency of 0.5Hz with an amplitude of 10V to the piezos, translating to roughly 22 microns. This was done first in the X axis, and then was stopped and repeated in the Y axis for each of the same laser powers we tested for Brownian motion. An example trace can be seen in [5]

It is worth observing that for lower laser power [6], the signals recorded are representative of a fixed bead on the slide, as discussed in the QPD calibration section. This may be due to the bead having fallen to the bottom of the trap, or it may simply be that at these powers the viscosity of the fluid overwhelms the laser. Most likely, it

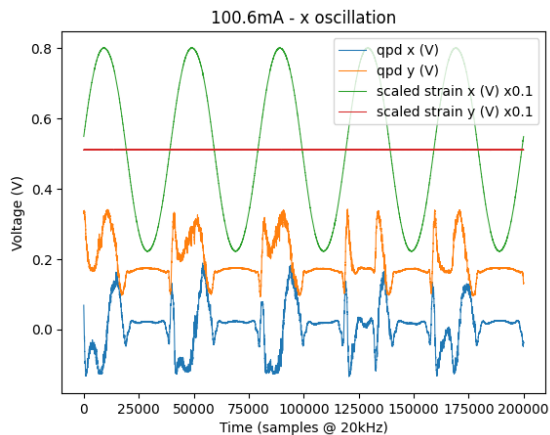


FIG. 6. XY scan at low laser current, showing traces that resemble a QPD calibration trace for fixed bead.

is a combination of these effects. However, this data was not analyzed as the corresponding QPD calibration data was not usable and so there was no way to transform it to position data in a timely manner. QPD calibration was later redone after discovering misalignment, but this process adjusted the position of the QPD in the apparatus and as a result the data would need to be retaken to avoid any bias.

V.3. QPD and discovering misalignment

Following the XY scans we set out to calibrate the QPD data. To do this, a new slide was assembled but with 1mol NaCl solution. This salinity causes the beads to stick to the glass slides, effectively fixing them. We then drive the slide stage such that we sweep out a grid pattern under the trap. From this, and knowing that passing the bead over the center of the trap should give the strongest scattering of the beam, we can deduce from the output QPD data when the bead has crossed over the center of the trap in one axis. We then take this data and plot stage strain in volts vs QPD in millivolts, and take the gradient to obtain a QPD to position conversion relation. This can then be applied as a product to QPD data say from the Brownian motion or the XY scans to produce correct position displacements.

Our initial attempts at QPD calibration produced data that confused us. Upon analysis, and noting that these changing QPD signals happened at the extremes of the sweeps in each axis, it was determined that the QPD was either misaligned, or the zeroing of the stage before the sweep was run had placed the bead in the corner of our measurement window just before we took data. As a result, we see QPD signals that represent what we might expect to see had we just clipped the edge of the bead before changing the direction of motion and leaving it again. See figure 7. The M and W shapes in the signal represent this partial bead coverage before reversing the

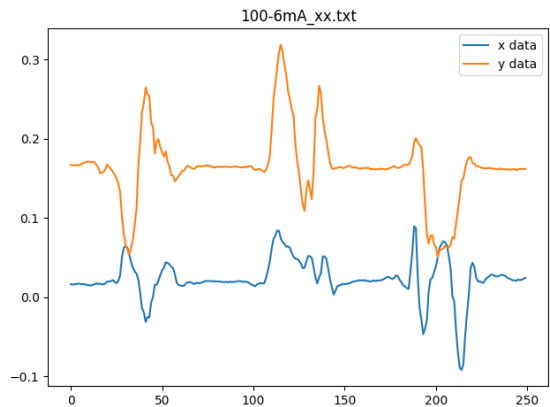


FIG. 7. Initial QPD data showing poor bead centering. The leftmost signal is as the sweep is zeroing, and passes over the center of the bead quite well. The proceeding signals showing M shapes do not pass through the bead sufficiently for good calibration.

stage. Interestingly, we can see a valid signal trace on the leftmost side, which is data from when the system is aligning to 0 coordinate and sweeping over Y to move the trap to the top of the window ready for x calibration. This data could be used to calibrate a y response.

QPD calibration was later performed again resulting in the conversion factor results for the QPD in figure 10. Here we see that the QPD response increases linearly with current supplied to the laser, which should be proportional to laser power. In our case, the Y response of the trap has a much steeper gradient. This could be due to trap asymmetry, but is also likely due to the stage not being perfectly level with normal to the z axis.

Of interest again are the results from the extremes of our laser power range. At 100mA, where we expect the behavior of the laser to be non linear we see a deviation away from the expected linear QPD response. At 400mA and above, we see a distorted QPD trace with asymmetric properties, indicating that the higher power of the beam is causing either deformation or slight movement of the bead as we pass through the trap. Because of this, future researchers may want to implement image based or other techniques to evaluate absolute trapped object position at high laser powers, or adjust error bars accordingly.

VI. FURTHER WORK

Following the obtaining of alpha and knowledge of how to calibrate a QPD well, it may be of interest to go forth and calibrate for 1 micron beads, opening up the field for DNA and E.Coli investigations. The organelles of an onion cell are also suitable for this scale and these experiments can provide motivation for further study.

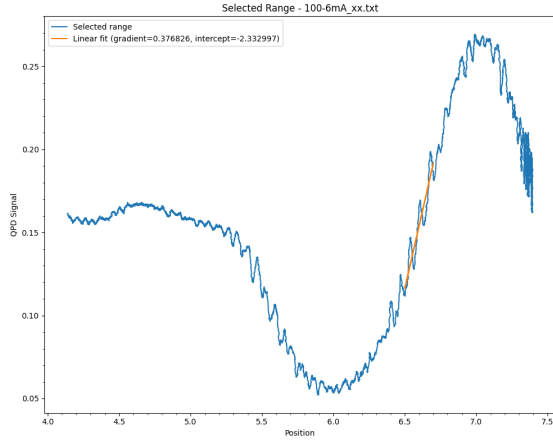


FIG. 8. Closeup showing a line fit to the gradient of a good QPD signal.

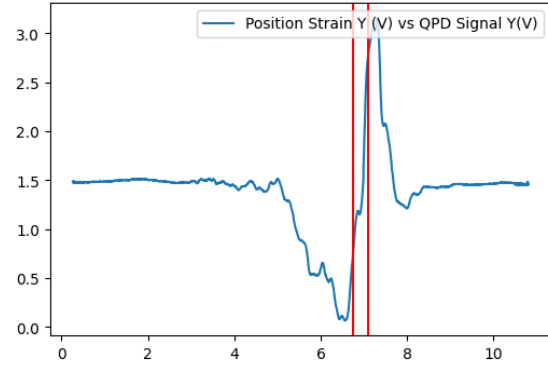


FIG. 9. QPD trace of a higher powered pass, showing asymmetry in the QPD signal

VI.1. Bibliography

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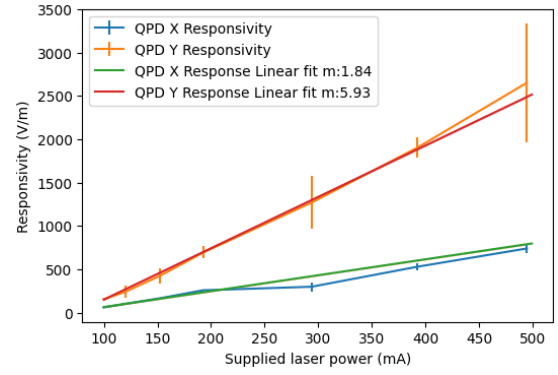


FIG. 10. Resulting QPD calibration from second pass, demonstrating a largely linear relationship between QPD response and laser power.

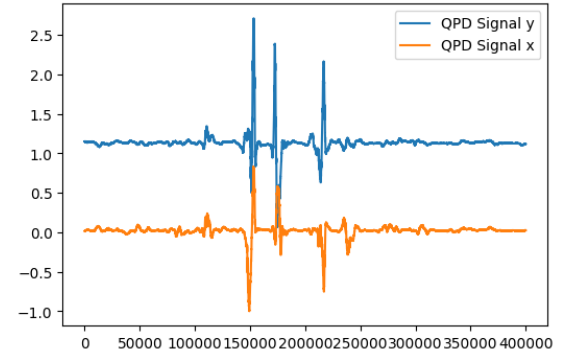


FIG. 11. An example of a decently done QPD data collection, with a clear sinusoid like shape in the signal trace.